Drift-Regularized Multi-Token Prediction via On-Policy Self-Distillation

Po-Wei Huang, Feng-Ting Liao
Department of Physics, University of Oxford
MediaTek Research

1 Objective

This research investigates latent representation alignment in policy distillation, aiming to improve multi-token generation beyond supervision based solely on token-level predictions. While conventional knowledge distillation primarily constrains output distributions, the geometry of intermediate hidden representations remains underdetermined.

We aim to develop a geometry-aware drifting regularizer for aligning student and teacher hidden-state distributions, and systematically compare its effects with paired feature regression. We investigate how these distinct alignment objectives influence multi-token rollout performance, including reasoning accuracy, effective prediction horizon, generation latency, and token throughput. In particular, we seek to identify the conditions under which population-level drifting or pointwise regression is more beneficial, and characterize their limitations across prediction horizons, regularization strengths, and model scales. The resulting analysis aims to establish practical guidelines for selecting latent-space supervision strategies that balance representation alignment, generation quality, and inference efficiency.

2 Novelty

We adopt the multi-token prediction (MTP) framework of Kirchenbauer et al. [6], which converts a pretrained autoregressive language model into a standalone multi-token predictor through on-policy self-distillation. The pretrained model serves as a frozen teacher, while a student initialized from the same checkpoint learns to predict multiple future tokens simultaneously. The teacher evaluates the student's proposed continuation autoregressively, providing token-level supervision without requiring an additional verification model during inference.

Our methodological contribution is to introduce drifting regularization [2] into the latent representation space of MTP self-distillation. Inspired by contrastive learning, drifting constructs an attraction–repulsion field using hidden representations from the teacher and student. Teacher representations provide positive samples that attract student states, while other student representations form a negative population that produces repulsive interactions. Unlike pointwise feature regression, drifting aims to align the student and teacher feature distributions without requiring exact one-to-one correspondence between their hidden states. We investigate whether this population-level objective better preserves the geometric structure of the teacher's representations across prediction horizons.

3 Impact

We hypothesize that improved latent representation alignment can enhance multi-token prediction beyond the gains achievable through token-level self-distillation alone. In particular, preserving the teacher's representation geometry may help the student maintain coherent and distinguishable predictive states across longer prediction horizons, potentially improving generation quality without sacrificing parallelism.

Preprint.

The central objective is to establish whether latent geometric alignment contributes to measurable improvements in both generation accuracy and inference efficiency. If successful, the proposed approach could enable the model to produce more reliable tokens per forward pass, thereby reducing generation latency and increasing token throughput under standalone MTP inference. Since drifting regularization is applied only during training, it introduces no additional inference module or verification stage.

More broadly, this study aims to clarify the relationship between representation geometry, token-level prediction quality, and the accuracy–acceleration trade-off in self-distilled language models. Such findings could inform the design of representation-level supervision for other parallel generation and model distillation frameworks.

4 Methods

4.1 On-policy MTP self-distillation

For training region r, let c_r denote a context prefix and k_r the number of future predictions. Following Kirchenbauer et al. [6], the flattened student input inserts $k_r - 1$ trainable `<MTP>` mask tokens. Repeated position IDs and a blocked causal attention mask allow active prediction states to attend to the prefix and preceding masks without altering copied context representations. At horizon j, the student produces hidden state $u^S_{r,j}$ and token distribution $p^S_{r,j}$. Its greedy proposal is $\hat{y}_{r,j} = \arg\max_v p^S_{r,j}(v)$.

The frozen teacher is evaluated on the student-forced continuation:

$$u^T_{r,j} = H_{\theta_T}(c_r, \hat{y}_{r,<j}), \qquad \bar{y}_{r,j} = \arg\max_v p_{\theta_T}(v \mid c_r, \hat{y}_{r,<j}). \tag{1}$$

With $N_{\text{act}} = \sum_r k_r$, the hard on-policy distillation loss is

$$\mathcal{L}_{\text{MTP}} = -\frac{1}{N_{\text{act}}} \sum_{r,j} \log p^S_{r,j}(\bar{y}_{r,j}). \tag{2}$$

For active index $i = (r, j)$, let $x_i = \phi_S(u^S_i)$ and $y^+_i = \phi_T(u^T_i)$ be features in a shared latent space, using the same normalization. Detached student features $y^-_l = \text{sg}(x_l)$ form the negative population, where sg denotes stop-gradient.

4.2 Latent regularization

We compare the unified objective

$$\mathcal{L} = \mathcal{L}_{\text{MTP}} + \lambda_R \mathcal{L}_R, \qquad R \in \{\text{SmoothL1}, \text{drift}\}. \tag{3}$$

The paired regression baseline uses SmoothL1 (Huber-type) regression:

$$\mathcal{L}_{\text{SmoothL1}} = \frac{1}{N_{\text{act}} d} \sum_{i=1}^{N_{\text{act}}} \sum_{m=1}^{d} \rho_\beta(x_{i,m} - \text{sg}(y^+_{i,m})), \tag{4}$$

where $\rho_\beta(z) = z^2/(2\beta)$ for $|z| < \beta$ and $\rho_\beta(z) = |z| - \beta/2$ otherwise. The implementation's β value will be reported with the final experimental configuration. For drifting, the feature-only Laplace kernel is

$$\kappa_h(x_i, y_l) = \exp\left(-\frac{\|x_i - y_l\|_2}{\tau_h}\right). \tag{5}$$

Positive and negative pools $\mathcal{I}^+_i$ and $\mathcal{I}^-_i$ contain teacher and detached student features, respectively. No prefix similarity term is used. Pool selection and any horizon grouping are separate from the kernel definition. For $s \in \{+, -\}$, define $\mathcal{J}^+_i = \mathcal{I}^+_i$ and $\mathcal{J}^-_i = \mathcal{I}^-_i \setminus \{i\}$. Normalize affinities as

$$a^s_{il} = \frac{\kappa_h(x_i, y^s_l)}{\sum_{m \in \mathcal{J}^s_i} \kappa_{\text{cond}}(i, m; y^s_m) + \epsilon}, \qquad l \in \mathcal{J}^s_i, \tag{6}$$

with $\epsilon > 0$ for numerical stability. The self-excluded drift estimate is

$$\widehat{V}_i = \sum_{l \in \mathcal{J}^+_i} a^+_{il}(y^+_l - x_i) - \sum_{l \in \mathcal{J}^-_i} a^-_{il}(y^-_l - x_i). \tag{7}$$

Table 1: Planned objective variants and their diagnostic roles.

ID	Objective	Diagnostic role
M0	Hard on-policy MTP	Token-only baseline
M1	M0 + paired latent SmoothL1	Pointwise alignment
M2	M0 + conditional full drift	Attraction and repulsion
M3	M0 + attraction-only drift	Repulsion ablation
M4	M0 + block-level full drift	Joint block coherence
M5	Soft on-policy KL distillation	Soft token supervision

Following the stopped-target construction of drifting models [2], the auxiliary objective is

$$\mathcal{L}_{\text{drift}} = \frac{1}{N_{\text{act}}} \sum_i \left\| x_i - \text{sg}\left(x_i + \eta \frac{\widehat{V}_i}{\|\widehat{V}_i\|_2 + \epsilon} \right) \right\|_2^2, \tag{8}$$

where η controls the target displacement. The block-level variant applies the same construction to concatenated normalized block features, $b_r^S = k_r^{-1/2} \,\text{vec}[x_{r,1}; \ldots; x_{r,k_r}]$, and their teacher counterparts, grouping blocks of the same length.

4.3 Controlled comparisons

The planned variants are summarized in Table 1. We will keep initialization, data, and training budgets comparable.The current strategy is to fix gradient-norm ratio between regularisation part and original MTP one to be approximately 0.30. All the studies below are freezed under this ratio to test under 1B Llama-3.1-Instruct model.

4.4 Evaluation and decision rules

Current 1B experiments train on MetaMathQA and evaluate on GSM8K using free-rollout generation. Additional benchmarks for larger models will be selected in a later phase, after establishing robust results on the present setup. We currently report GSM8K answer accuracy and the realized average number of emitted tokens per forward pass. The frozen teacher under sequential decoding ($k = 1$) provides the capability-retention reference. All model-selection decisions will use held-out validation data.

We will measure token-level expected calibration error (ECE), following the calibration perspective of Joshi et al. [5]. Separately, we will compare teacher and student feature populations using sliced Wasserstein-2, maximum mean discrepancy [4], Sinkhorn transport cost [1], and 1-nearest-neighbor two-sample accuracy, with complementary distributional diagnostics motivated by Esteban-Casadevall et al. [3]. Features will use shared normalization and any PCA projection fitted only on training features. Diagnostics will be reported per horizon; nearest-neighbor accuracy approaching 50% indicates reduced population distinguishability under balanced sampling.

We will evaluate both static and confidence-adaptive decoding. Realized token acceleration is

$$k_{\text{eff}} = \frac{\text{emitted non-padding tokens}}{\text{decoding forward passes}}, \tag{9}$$

which will be reported alongside measured throughput and end-to-end latency. For a single operating-point summary, validation will select the fastest setting within one percentage point of the sequential teacher's validation accuracy, when such a setting exists; it will then be frozen for test evaluation. We will report uncertainty over evaluation examples and, where feasible, training seeds. If SmoothL1 outperforms drift, or token-only training outperforms both regularizers, we will report that outcome and examine sampling noise, regularization strength, and conditioning rather than assume the proposed ordering holds.

5 Risks

Noisy or mismatched feature populations. Conditional drift depends on the size and relevance of local teacher and student pools. Sparse horizon groups or unrelated contexts may produce noisy

updates. We will monitor pool sizes, kernel affinities, feature norms, and per-horizon diagnostics; cap or subsample pools for tractability; and retain paired SmoothL1 and attraction-only drift as diagnostic alternatives. Updates with insufficient pool support will be skipped.

Overregularization and misleading alignment. Forcing feature alignment may reduce token accuracy or encourage feature collapse. Matched auxiliary gradient strength and coefficient sweeps will help separate the effect of the regularizer from loss scaling. Lower latent distances or ECE alone will not be treated as evidence of improved reasoning: task scores and the teacher-retention gap remain necessary checks.

Training overhead and deployment trade-offs. Pairwise kernel interactions increase training compute and memory, and a larger configured block size need not yield proportional wall-clock speedup. We will bound population sizes and profile teacher passes, kernel computation, and training memory. Deployment comparisons will use identical hardware, batch sizes, prompt lengths, and generation limits.

Generalization and evaluation validity. Training on mathematical traces may degrade other capabilities, motivating the broader evaluation suite. Validation and test splits will remain separate, and training data will be checked for benchmark overlap. Faster generation may also amplify existing model errors; capability retention and confidence calibration will be assessed before making practical deployment claims.

6 Timeline & Preliminary Results

The project has completed baseline reproduction and is currently transitioning from drifting pilot experiments to systematic ablations. Dates and durations for the remaining phases depend on compute availability.

1. **Baseline reproduction (completed):** establish the on-policy MTP pipeline, sequential teacher reference, paired SmoothL1 baseline, and evaluation harness.
2. **Drift implementation and 1B pilot (in progress):** implement conditional pools, self-exclusion, and stopped-target updates; profile memory and runtime; Adjusting regularisation strength will be operated under the actions of running 8B Llama model. Under this intermediate step, proving drifting effectively gains advantages over baselines is the main objective.
3. **Main experiments and ablations (next):** run the block-size and confidence-threshold sweeps, compare all objective variants, and collect task, calibration, and latent-alignment measurements.
4. **Analysis and reporting (planned):** quantify uncertainty, analyze failure cases and capability retention, and prepare reproducible configurations and the final manuscript.

6.1 Current Progress (October 2026)

The on-policy MTP baseline has been reproduced, and pilot comparisons of unregularized MTP, SmoothL1, drifting, and combined objectives have been performed at the 1B scale. The following observations are preliminary; plots summarize the supplied experimental runs rather than a completed multi-seed significance study. The experimental configurations are summarized in Table 2. We compare hard-teacher MTP self-distillation against drifting regularization, paired SmoothL1 feature regression, and their combination. Additional ablations such as kernel bandwidth selection and the maximum training prediction horizon $k_{\max}$ will be investigated in 8B models.

Feature dispersion across horizons. Figure 1 reports the student-to-teacher feature spread ratio, defined as the mean pairwise distance between distinct student hidden states divided by the corresponding mean distance between teacher states. A ratio below unity indicates a more contracted student distribution, while a ratio above unity indicates greater dispersion. SmoothL1 systematically contracts student representations, with the effect becoming more pronounced at longer prediction

Table 2: Experimental configurations and corresponding figure legends. Auxiliary weights (w) are loss coefficients, while percentages indicate approximate auxiliary-to-MTP gradient-norm ratios. All experiments use hard-teacher on-policy MTP cross-entropy as the primary training objective.

Figure legend	**Training configuration**
`nodrift`	MTP loss only, using hard-teacher cross-entropy supervision.
`drift selfpos w0.05`	MTP loss with self-positive drifting regularization ($w_{\text{drift}} = 0.05$), corresponding to an approximately 30% auxiliary-to-MTP gradient-norm ratio.
`drift selfpos w0.05 tau0.02-0.1-0.2`	MTP loss with self-positive drifting regularization ($w_{\text{drift}} = 0.05$; approximately 30% gradient-norm ratio) and adjusted kernel bandwidths ($\tau = 0.02, 0.1, 0.2$).
`nodrift kmax5`	MTP loss only, with the maximum training prediction horizon restricted to $k_{\max} = 5$.
`smoothl1 w0.53`	MTP loss with paired SmoothL1 feature regression ($w_{\text{SmoothL1}} = 0.53$), corresponding to an approximately 20% auxiliary-to-MTP gradient-norm ratio.
`drift selfpos w0.05 + smoothl1 w0.53`	MTP loss with both self-positive drifting ($w_{\text{drift}} = 0.05$; approximately 30%) and paired SmoothL1 regression ($w_{\text{SmoothL1}} = 0.53$; approximately 20%).

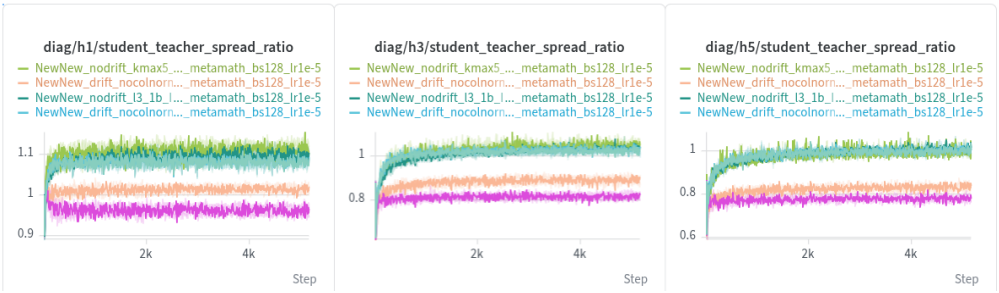


Figure 1: Student-to-teacher hidden-feature spread ratios over training steps for horizons 1, 3, and 5. Values near one indicate similar measured spread, not necessarily identical feature distributions. Run names are preserved from the experiment dashboard.

horizons. In contrast, drifting better preserves the teacher's feature spread, suggesting that population-level alignment can maintain relative geometric structure without forcing individual student states to collapse toward their paired teacher targets.

Distributional diagnostics. Figure 2 compares MMD and Sinkhorn distance across prediction horizons. Drifting configurations generally yield lower values than the unregularized baseline in the plotted range. SmoothL1 is competitive and lower than drifting in terms of Sinkhorn distance, as the pair distance between student and teacher is the goal for minimisation under smoothL1 training purpose, which is the quantity optimal transport values the most. MMD which cares about about two individual spread of groups will capture how alignment two generated distributions are. The results shows that drifting chemem has better alignments across MTP region than smoothL1 one, whereas under NTP(k=1) region smoothL1 still performs the best. Both metrics generally worsen as the horizon increases.

Free-rollout accuracy and acceleration. Figure 3 reports GSM8K single-shot chain-of-thought accuracy against the average effective k from free generation. Solid curves represent confidence-adaptive decoding and dashed curves fixed-k settings. At some matched acceleration values, SmoothL1 and drifting runs outperform the unregularized model, but no single objective dominates throughout the displayed range. The reduced-$k_{\max}$ run shows encouraging accuracy in part of the adaptive-decoding regime, although this does not establish a uniform gain from decreasing $k_{\max}$.

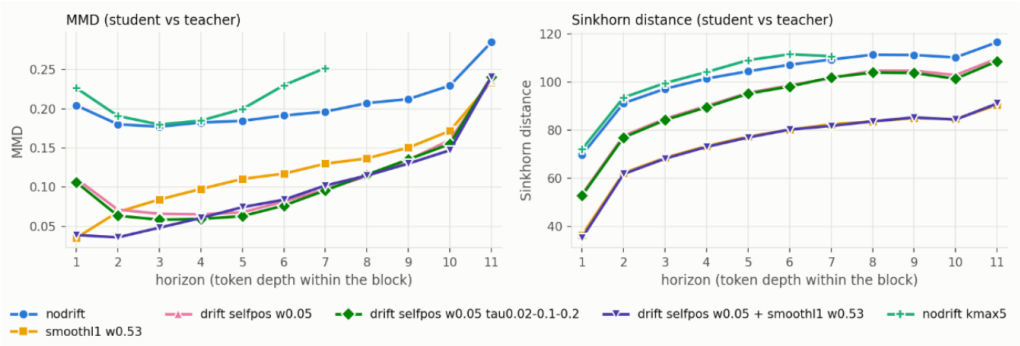


Figure 2: Held-out student–teacher maximum mean discrepancy (left) and Sinkhorn distance (right) as a function of prediction horizon. Lower is better for both diagnostics, but neither directly measures task accuracy. The different objectives exhibit horizon-dependent trade-offs.

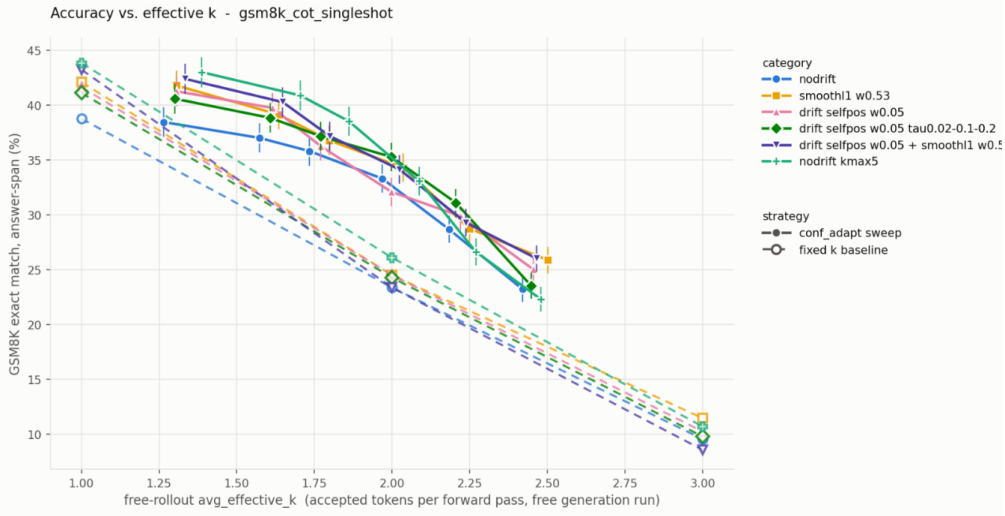


Figure 3: GSM8K free-rollout answer accuracy versus realized effective k for the 1B pilot runs. Solid lines show confidence-adaptive sweeps; dashed lines show fixed-k results. Higher accuracy and larger effective k are preferred, and comparison should be made at matched operating points rather than by configured $k_{\max}$ alone.

Next steps. We will test the stability of these observations under matched training budgets and repeated seeds, quantify paired feature errors separately from distribution distances, and examine whether horizon-specific feature geometry predicts improvements in free-rollout accuracy. We will also study the effect of $k_{\max}$ and expand to larger checkpoints and broader benchmarks once the 1B and 8B methodology are consolidated.

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